

Does the Production Approach Recover Markup Changes over Time?*

Paolo Mengano
Esade Business School

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Abstract

Market power is a fundamental element of the economy, and markups, defined as price over marginal cost, are its most common measure. Recent advances show that markup levels are difficult to estimate from production data, while changes over time are considered easier to recover. Leveraging a unique dataset, I test this claim by comparing estimated annual markup changes with firms' survey responses on the evolution of the margins they charge over costs. Production-based estimates overstate the share of firms with rising markups by roughly 50 percentage points, and estimated and reported directions disagree for 57% of firms. These findings cast doubt on markup dynamics recovered with the production approach.

Keywords: Markups, Firm Dynamics, Production Functions, Firm Surveys.

JEL codes: C33, D24, L11, L25.

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1 Introduction

Recent evidence on the rise in aggregate markups in the United States (De Loecker et al., 2020) and in the rest of the world (De Loecker and Eeckhout, 2018; Díez et al., 2021) has put market power at the center of the academic debate. This surge in market power has indeed been associated with the secular decline in the labor share (Autor et al., 2020), declining business dynamism (De Loecker et al., 2021), aggregate fluctuations (Burstein et al., 2025), innovation (De Ridder, 2024), and consumer welfare (Edmond et al., 2023). Yet, markups—defined as price over marginal cost—are an economic object that is not directly observed in the data. Most of this evidence therefore relies on a structural method, the production approach, which recovers markups from the cost-minimization condition for a flexible input, combining an estimated output elasticity with the input’s expenditure share (Hall, 1988; De Loecker and Warzynski, 2012). Existing literature points to several challenges in applying this approach to financial information, both methodological, such as the identification of production functions, and empirical, such as the use of revenue rather than physical output and the choice of flexible input (Gandhi et al., 2020; Bond et al., 2021; Raval, 2023; Kirov et al., 2026). While these challenges are widely recognized for measuring markup levels, changes and time trends are often considered more reliable (De Ridder et al., 2026). As with any study measuring an unobservable component, however, there are few direct ways to validate the results. Existing assessments are therefore predominantly methodological, rely on simulated data, or use indirect validation by examining whether estimates reproduce selected economic relationships.

This paper addresses these challenges by providing the first direct comparison between production-based markup changes and independently elicited survey evidence on margin dynamics. I use the EU-EFIGE database, which combines a representative survey of European manufacturing firms with their balance sheet information. The survey first asks firms how they set their prices, allowing me to distinguish between price-takers and firms that actively set prices by applying a margin over costs. Non-price-taker firms are subsequently asked whether the size of their margin increased, decreased, or remained constant during the previous year. I compare this response with the change in a production-based markup estimated from the same firm’s accounts and over the same period. The two measures are systematically at odds: reported and estimated directions differ for more than half of firms, casting serious doubt on whether the production approach reliably recovers annual markup changes.

The survey does not directly observe the markup over marginal cost. Firms may calculate their margins over total or variable costs, and the questionnaire does not specify which expenditures should be included in either category. Nevertheless, the response provides information on the evolution of a margin explicitly connected to

the firm's pricing strategy and collected independently from the production function and accounting expenditure share used to estimate markups. This makes it possible to assess whether production-based markup changes recover which firms experienced an improvement in the price–cost margins used to set their prices.

During the onset of the Global Financial Crisis, only 11% of firms report that their margin increased between 2008 and 2009. By contrast, markups are unobservable. For a cost-minimizing firm, the production approach measures them as the ratio of an economic term, i.e. the output elasticity of materials, to an accounting term, i.e. the revenue share of materials, for a cost-minimizing firm. Starting from the accounting component of the production approach, I find that the materials margin increases for 70.1% of firms. Inferring markup dynamics from changes in the materials expenditure share would therefore overestimate the share of firms with increasing margins by 59 percentage points relative to the survey evidence. I then estimate a flexible translog production function using a control function approach to allow output elasticities to vary across firms and over time. Introducing estimated output elasticities reduces the share of markup increases to 60.9%. The resulting difference with the survey remains 49.9 percentage points.

The disagreement is also present when the two changes are compared within the same firm. Reported and estimated directions agree for only 43% of the sample. The largest group consists of firms reporting that their margin did not increase while receiving an estimated markup increase. These firms account for 53.4% of the weighted sample. Moreover, although estimated markup increases are somewhat more frequent among firms reporting higher margins, the magnitude of estimated markup growth is essentially unrelated to the survey response in the linear regressions. The result is not driven by the classification of small changes around zero and is not concentrated in a particular segment of the distributions of markup levels, markup changes, sales, or materials expenditure. Hence, the disagreement appears to be a general feature of the production approach rather than a problem confined to a particular group of firms.

I next consider whether firms' pricing strategies can explain these findings. If the disagreement originates from the fact that reported margins include fixed costs, the comparison should improve among firms pricing over variable costs. It does not. Among total-cost pricers, estimated markups increase for 60.5%, while only 11.4% report a margin increase. Among variable-cost pricers, the corresponding shares are 62.6% and 9.9%, producing an even larger difference of 52.7 percentage points. Estimated markup growth is also unrelated to reported margin changes within both groups. Selecting firms whose pricing strategy is closest to the flexible-input approach therefore does not improve the comparison.

Finally, I examine whether fixed costs can reconcile estimated and reported changes among total-cost pricers. Following De Ridder (2024), I recover the fixed cost share

from the difference between the margin implied by the estimated markup and the operating-profit margin. I then hold the fixed cost level recovered in 2008 constant in real terms and allow its incidence over revenues to change in 2009. This adjustment reduces the incidence of increases from 60.5% for estimated markups to 54.8% for adjusted margins and raises agreement with the survey from 43.9% to 49.3%. Among firms reporting a non-increase but receiving an estimated markup increase, fixed cost absorption reverses the estimated direction in 11.4% of cases. The adjusted increase rate nevertheless remains 43.4 percentage points above the rate reported by firms. Under this benchmark, fixed cost absorption reduces the gap but leaves most of it unexplained.

These findings raise serious concerns about what annual production-based markup changes can recover from standard firm-level data. This limitation matters for event studies, within-firm decompositions, and other applications that investigate drivers and consequences of changes in estimated markups. These changes feed directly into pass-through regressions, inflation decompositions, and merger retrospectives; when the firm-level variation provides little information, those exercises inherit the problem. It also raises questions about aggregate trends constructed from the same annual firm-level variation.

This paper contributes to the literature on productivity estimation and markup measurement using the production approach. A large literature studies the conditions under which markup levels can be recovered from production data (De Loecker et al., 2016; Gandhi et al., 2020; Bond et al., 2021; De Loecker and Syverson, 2021), while recent papers develop alternative estimators or assess whether standard estimates remain informative when output prices are unavailable (Kirov et al., 2026; De Ridder et al., 2026). Much less is known about what estimated changes capture. Raval (2023) shows that markups constructed from different flexible inputs can be negatively correlated and imply opposing trends. Conversely, De Ridder et al. (2026) shows that revenue-based estimates may preserve useful information about markup variation—revenue- and quantity-based markups correlate at 0.80 in first differences in their French data—, while Forsbacka (2026) finds that production-based estimates reproduce the relationship between markups and local market concentration. These validations remain internal to the production framework, as estimates are checked against other estimates built from the same first-order condition. Detailed evidence on prices and costs is available in more narrowly defined settings: Atkin et al. (2017) collect production-line data for Egyptian rug producers, including detailed information on product specifications and quality. This paper complements these studies by comparing estimated changes with information collected outside the production function calculation for the same firms and over the same period.

The paper also relates to a long literature using surveys to understand how firms set prices (Blinder et al., 1998; Fabiani et al., 2007). Recent work shows that firms

frequently price over average or accounting costs rather than over the marginal costs appearing in standard models (Altomonte et al., 2015; Gutmann and Sangani, 2026). I use this information differently. Rather than treating firms' stated pricing strategies as a model of optimal price setting, I use the reported evolution of their margins to assess the empirical content of production-based markup changes. The persistent disagreement indicates that estimated annual markup changes do not track the margins firms report, and that further validation against independent information on firms' pricing and costs is needed.

2 How Did Margins Evolve?

I use survey information from EFIGE to measure the evolution of firms' margins and compare it with markup changes estimated from the same firms' accounts. I first describe the database and the survey questions, and then discuss how the reported changes can be used to study markup dynamics.

2.1 The EFIGE Database

I use the EU-EFIGE/Bruegel-UniCredit database (hereinafter, the EFIGE database), which combines a representative survey of manufacturing firms with firm-level balance sheet information. The survey covers firms with at least ten employees in Austria, France, Germany, Hungary, Italy, Spain, and the United Kingdom.¹ Developed under the EFIGE project—supported by the European Commission—the database is designed to capture the manufacturing structure of each country through a stratified sampling approach by industry, region, and firm size. Large firms, defined as firms with more than 250 employees, are oversampled, and the database provides sampling weights that account for this design.

The EFIGE survey was conducted between January and May 2010 using a standardized questionnaire common to all countries.² The questionnaire is organized in six modules covering firm structure and ownership, workforce composition, investment and innovation, internationalisation, finance, and markets and pricing. Most questions refer to firms' activities in 2008, while selected questions record changes extending into 2009. The latter include the questions on pricing rules and margin changes used in this paper.

The survey was subsequently matched with balance sheet information from the Amadeus database by Bureau van Dijk covering the period 2001–2014. The linked

¹Austria and Germany are excluded from the analysis due to the insufficient number of observations (Békés et al., 2016).

²The questionnaire was administered via CATI (Computer Assisted Telephone Interview) or CAWI (Computer Assisted Web Interview) procedures. The complete questionnaire is available on the Bruegel website: <https://www.bruegel.org/dataset/efige>.

accounts allow me to estimate production functions and construct annual markup changes for the surveyed firms. Altomonte and Aquilante (2012) show that weighted statistics from EFIGE align closely with corresponding aggregate statistics.³

Appendix Table A.1 reports summary statistics for the sample of analysis.

2.2 Margins in Survey Data

I draw from the markets and pricing module of the EFIGE survey to study the evolution of firms' margins. Question E10 first asks firms how they set their prices. Respondents can report that prices are set as a margin over total costs, as a margin over variable costs, by reference to prices prevailing in the market, through regulation, or using another method. These answers allow me to distinguish between price-takers and firms that actively set their prices.⁴ Firms reporting that they price over either total or variable costs subsequently receive question E11: "During the last year, the size of your margin: increased, decreased, [or] was equal?"

The sequence of the two questions makes the survey particularly useful for studying markup dynamics. Firms are not asked whether their profitability or general economic conditions improved. They are asked about the evolution of a margin explicitly connected to a pricing rule in which they actively set prices relative to costs. Among firms providing a valid response to E10, 57.9% are active price setters, while 38.5% are price-takers. The share of active price setters varies substantially across countries, ranging from 32.8% in Hungary to 66.8% in Spain. Appendix Table A.2 reports the corresponding shares for every country.

I classify active price setters according to whether they report an increase in their margin. The resulting indicator equals one if the answer to E11 is "increased" and zero if the margin decreased or remained unchanged. Only 11% of firms in the final comparison sample report that their margin increased. This low increase rate is consistent with the timing of the survey, which was conducted shortly after the sharp contraction in economic activity associated with the Global Financial Crisis. The period is also a relevant test: large shocks are precisely when researchers turn to firm-level markup changes, in event studies, crisis episodes, and pass-through analyses. The responses, moreover, track realized outcomes: broader accounting margins constructed from the same firms' accounts—adding labor costs, depreciation, or all operating expenses—are substantially more likely to fall among firms reporting that their margin did not increase.⁵

³The EFIGE database has been used in numerous papers, including Altomonte et al. (2013, 2016); Steinberg (2019); Pellegrino and Zingales (2017); Ferri et al. (2019).

⁴Throughout the paper, I define non-price-takers, or active price setters, as firms that explicitly report setting prices by applying a margin over total or variable costs. Firms reporting that their prices are determined by prices prevailing in the market or are subject to regulation are classified as price-takers. Other pricing rules are not assigned to either group.

⁵Among firms reporting a margin non-increase, the EBITDA margin does not increase for 60.2%

Because E11 refers to changes “during the last year,” I compare the survey response with the change in accounting variables and estimated markups between 2008 and 2009. This period provides the closest correspondence between the wording of the question and the available annual accounts. Appendix Table B.3 considers alternative timing conventions.

2.3 What Do Reported Margins Imply for Markups?

I now turn to what the survey evidence implies for markup dynamics. Let $P_{it} > 0$ denote the firm’s price and $C_{it} > 0$ the unit cost used to calculate its margin. Define the margin and the corresponding price–cost ratio as

$$m_{it} = \frac{P_{it} - C_{it}}{P_{it}} = 1 - \frac{1}{\mu_{it}^C}, \quad \mu_{it}^C = \frac{P_{it}}{C_{it}}. \quad (1)$$

The margin is strictly increasing in the corresponding price–cost ratio, thus changes in the margin and the price–cost ratio have the same sign.⁶

The EFIGE questionnaire does not provide a formal definition of margins nor require firms to calculate their margins using marginal cost. It identifies whether firms price over total or variable costs, but it does not specify which expenditures are included in either category or whether costs are measured at the product or firm level. The cost concept used by the firm may therefore differ from the marginal cost recovered by the production approach.

Consider first a firm pricing over variable costs. Let $VC_{it}(Q)$ denote its variable-cost function and define average variable and marginal cost as

$$AVC_{it} = \frac{VC_{it}(Q_{it})}{Q_{it}}, \quad MC_{it} = \frac{\partial VC_{it}(Q_{it})}{\partial Q_{it}}. \quad (2)$$

The output elasticity of variable costs satisfies

$$\eta_{it}^C = \frac{\partial \log VC_{it}(Q_{it})}{\partial \log Q_{it}} = \frac{MC_{it}}{AVC_{it}}, \quad \mu_{it}^{MC} = \frac{\mu_{it}^{AVC}}{\eta_{it}^C}, \quad (3)$$

where $\mu_{it}^{AVC} = P_{it}/AVC_{it}$ and $\mu_{it}^{MC} = P_{it}/MC_{it}$. It follows that

$$\Delta \log \mu_{it}^{MC} = \Delta \log \mu_{it}^{AVC} - \Delta \log \eta_{it}^C. \quad (4)$$

of firms, against 37.1% among firms reporting an increase—a difference of 23.1 percentage points ($p < 0.01$). The corresponding difference for the materials margin, defined as one minus the materials share of revenue, is 6.1 points.

⁶An alternative convention expresses the price–cost wedge relative to costs, $\tilde{m}_{it} = (P_{it} - C_{it})/C_{it} = \mu_{it}^C - 1$. This measure is also strictly increasing in μ_{it}^C and therefore produces the same implications for the sign of changes.

Average variable cost and marginal cost can clearly be different in many settings. Their associated price–cost ratios nevertheless have the same proportional changes whenever the output elasticity of variable costs is constant. A sufficient condition is that the production function is homogeneous of a time-invariant degree in the variable inputs, holding fixed inputs constant.

Under cost minimization and competitive input markets, variable costs then take the form $VC_{it}(Q) = \kappa_{it}Q^{\eta_i}$, where η_i is the reciprocal of that degree. Marginal cost is proportional to average variable cost, $MC_{it} = \eta_i AVC_{it}$. A change in productivity shifts both cost curves through κ_{it} without changing this proportionality. The margin over average variable cost and the markup over marginal cost therefore move in the same direction without requiring marginal cost to be constant. Average variable and marginal cost may both increase or both decrease with output, depending on the curvature of the cost function.

A constant elasticity is plausible over the one-year horizon of the comparison. The condition concerns the shape of the variable cost function rather than its level: productivity and input-price movements shift κ_{it} , while η_i changes only when the production technology itself changes. Such changes are infrequent. Even over the three-year window 2007–2009, 54.6% of firms in the comparison sample report no process innovation—defined by the survey as the adoption of a production technology that is new or significantly improved—and the median firm invested 5% of turnover per year in plants, machinery, equipment, and ICT. Among variable-cost pricers, for whom the mapping is most direct, 55.4% report no process innovation over the same window.

The mapping requires an additional term for firms pricing over total costs. Let F_{it} denote fixed costs, let $ATC_{it} = AVC_{it} + F_{it}/Q_{it}$ denote average total cost, and let $R_{it} = P_{it}Q_{it}$ denote revenues. The margin over total costs can be written as

$$m_{it}^{\text{TC}} = 1 - \frac{ATC_{it}}{P_{it}} = 1 - \frac{1}{\eta_{it}^C \mu_{it}^{\text{MC}}} - \frac{F_{it}}{R_{it}}. \quad (5)$$

For total-cost pricers, a sufficient condition for the margin and the markup to move in the same direction is that both the output elasticity of variable costs and the fixed cost share of revenues remain constant. When fixed operating expenditures do not change, a decline in revenues raises this share and can cause the total-cost margin to fall even if the markup increases. The fixed cost adjustment in Section 4 examines this possibility.

These relationships provide the basis for the empirical comparison. Under these conditions, if the production approach accurately recovers markup changes, estimated increases should be concentrated among firms reporting increasing margins. Moreover, larger estimated increases should predict a higher probability that the survey reports an increase. The comparison is most direct for variable-cost pricers, for whom the mapping depends on the stability of the output elasticity of variable costs. For total-cost

pricers, it also depends on fixed costs. I first examine the general correspondence between reported and estimated changes and then use the distinction between pricing strategies in Section 4 to examine whether total-cost pricing and fixed costs can explain the disagreement.

3 Markup Dynamics from Production Data

3.1 The Production Approach

Markups are defined as the ratio between output price and marginal cost, neither of which is separately observed in standard firm-level accounting data. I estimate them using the production approach originally proposed by Hall (1988) and developed for firm-level applications by De Loecker and Warzynski (2012). The approach recovers the markup from a firm's expenditure on an input that can be adjusted freely when production decisions are made.

Consider a firm producing output Q_{it} using a flexible input V_{it} and a collection of inputs that cannot be adjusted freely within the period. Let P_{it}^V denote the price of the flexible input and λ_{it} the marginal cost of production. Cost minimization implies

$$P_{it}^V = \lambda_{it} \frac{\partial Q_{it}}{\partial V_{it}}. \quad (6)$$

Multiplying both sides by V_{it}/Q_{it} and using $\mu_{it} = P_{it}/\lambda_{it}$ gives

$$\mu_{it} = \frac{\theta_{it}^V}{\alpha_{it}^V}, \quad \alpha_{it}^V = \frac{P_{it}^V V_{it}}{P_{it} Q_{it}}, \quad (7)$$

where θ_{it}^V is the output elasticity of the flexible input and α_{it}^V is its expenditure as a share of revenue. The expenditure share is directly observed in the accounts. The output elasticity must instead be recovered from an estimated production function.

I use materials as the flexible input. Materials expenditure can generally be adjusted with production within the year and is commonly used for this purpose in applications of the production approach (De Loecker and Warzynski, 2012; De Loecker et al., 2020). Employment is less plausibly treated as freely adjustable, especially in European countries. Hiring, dismissal, and organizational costs make labor adjustment costly, while employment protection further slows the response of employment to output in European manufacturing.⁷ Labor and capital therefore enter the production function, but materials provide the first-order condition used to recover the markup.

I estimate a gross-output production function using deflated revenue as output,

⁷See, for example, Bentolila and Bertola (1990), Abraham and Houseman (1993), Hamermesh and Pfann (1996), and Mengano (2025).

tangible fixed assets as capital, the number of employees as labor, and deflated materials expenditure as the flexible input. Let y_{it} , k_{it} , l_{it} , and x_{it} denote the logarithms of these variables, respectively, where x_{it} denotes log materials. The production function is

$$y_{it} = f_c(k_{it}, l_{it}, x_{it}; \beta_c) + \omega_{it} + \varepsilon_{it}, \quad (8)$$

where ω_{it} is productivity observed by the firm when choosing its inputs, ε_{it} is a transitory shock, and the subscript c indicates that the parameters are estimated separately for each country.

I follow the two-stage control-function approach of Akerberg et al. (2015), building on Olley and Pakes (1996) and Levinsohn and Petrin (2003). The first stage uses a flexible polynomial in capital, labor, and materials to recover expected output net of the transitory shock. It also includes year effects and the firm's survey-weighted share of sales within its country, industry, and year. In the second stage, productivity is set to follow a first-order Markov process, and the production function parameters are identified using moment conditions based on current capital and lagged variable inputs. Appendix B reports the complete estimating equations and moment conditions, while Appendix Table B.1 summarizes the country-specific materials elasticities.

I use a translog specification, which allows the materials elasticity to vary across firms and over time. With x_{it} denoting log materials, the estimated elasticity is

$$\widehat{\theta}_{it}^M = \widehat{\beta}_c^M + 2\widehat{\beta}_c^{MM}x_{it} + \widehat{\beta}_c^{KM}k_{it} + \widehat{\beta}_c^{LM}l_{it}. \quad (9)$$

Combining this elasticity with the observed materials expenditure share produces the firm-year markup estimate.

3.2 Reported and Estimated Markup Dynamics

The markup formula in Equation (7) shows that estimated markup changes can originate from either the estimated output elasticity or the observed materials expenditure share. Taking changes gives

$$\Delta \log \widehat{\mu}_{it} = \Delta \log \widehat{\theta}_{it}^M - \Delta \log \alpha_{it}^M. \quad (10)$$

If the output elasticity were constant within firms, an estimated markup would increase if and only if the materials expenditure share fell. Equivalently, define the materials margin as

$$m_{it}^M = 1 - \alpha_{it}^M. \quad (11)$$

The materials margin can be calculated directly from the accounts and isolates the variation in estimated markups coming from the accounting denominator.

Table 1 begins with this comparison. Only 11% of firms report that their margin

increased between 2008 and 2009. By contrast, the materials margin increases for 70.1% of firms. Inferring markup dynamics from the materials expenditure share would therefore overestimate the share of firms with increasing margins by 59 percentage points relative to the survey evidence.

Allowing output elasticities to vary across firms and over time reduces the gap but does not resolve it. Once the estimated materials elasticity is included, markups increase for 60.9% of firms. Changes in the estimated elasticity therefore reduce the share of increases by approximately nine percentage points. The estimated increase rate nevertheless remains 49.9 percentage points above the share reported in the survey.

The decomposition in Equation (10) shows why the gap remains so large. Among firms reporting that their margin did not increase, 76% also have an estimated materials elasticity that does not increase. Nevertheless, estimated markups rise for 66% of this group. For these firms, the decline in the materials expenditure share more than offsets the movement in the estimated elasticity and generates an estimated markup increase.

Estimating the production function reduces the disagreement but does not change what it implies. During a year in which almost nine out of ten firms report that their margins did not increase, the production approach assigns a markup increase to more than six out of ten firms. The next subsection examines whether this aggregate gap is also present when reported and estimated changes are compared within the same firm.

3.3 From Aggregate to Firm-Level Dynamics

The large difference in the incidence of increases establishes substantial disagreement between the two measures. Reported margin increases could nevertheless be concentrated among firms receiving estimated markup increases. I examine this possibility by comparing the reported and estimated directions for the same firm.

The two directions agree for only 43% of the sample, and the disagreement is highly asymmetric. The largest group consists of firms reporting that their margin did not increase while receiving an estimated markup increase. These firms account for 53.4% of the weighted sample. The opposite case—a reported margin increase accompanied by an estimated markup non-increase—accounts for only 3.5%. The disagreement reflects a systematic tendency of the production approach to assign increases to firms reporting no improvement in their margins.

Estimated markup increases are somewhat more frequent among firms reporting higher margins. Markups rise for 68% of firms reporting an increase and for 60.1% of firms reporting a decrease or no change. The difference is 8 percentage points, with a p -value of 0.008. Although this difference has the expected sign, it is small relative to the unconditional estimated-increase rate. The production approach assigns an increase to the majority of firms in both survey groups and therefore provides little separation

between them.

I next examine whether the magnitude of estimated markup growth predicts the survey response. I estimate the survey-weighted linear probability model

$$\mathbb{1}\{\text{reported margin increase}_i\} = \beta_0 + \beta_1 \Delta \log \hat{\mu}_i + u_i. \quad (12)$$

If estimated markup changes recover the evolution reported by firms, β_1 should be positive: firms receiving larger estimated increases should be more likely to report that their margins increased.

The estimated coefficient is 0.008, with a p -value of 0.774. Estimated log markup growth is therefore essentially unrelated to the survey response in this linear specification. Appendix Table B.4 shows that this finding is robust in specifications with country and industry fixed effects and controls for firm size.

Taken together, the incidence, within-firm, and continuous comparisons indicate that estimated changes provide little information about which firms improved their margins. The next section examines whether differences in the costs used by firms and by the production approach can explain this result.

Table 1: Reported margins, materials shares, and estimated markups

	Estimate	Standard error
<i>Panel A. Share of firms with an increase</i>		
Reported margin	11.04	0.63
Materials margin	70.07	0.91
Estimated markup	60.93	0.97
Materials margin minus reported margin	59.03***	1.08
Estimated markup minus reported margin	49.90***	1.13
<i>Panel B. Increase rates conditional on the reported direction</i>		
Materials margin: reported non-increase	69.43	0.97
Materials margin: reported increase	75.24	2.59
Difference: increase minus non-increase	5.81**	2.77
Estimated markup: reported non-increase	60.05	1.03
Estimated markup: reported increase	68.02	2.82
Difference: increase minus non-increase	7.96***	3.00
<i>Panel C. Predicting a reported margin increase</i>		
Log estimated-markup growth: all firms	0.008	0.029
Log estimated-markup growth: total-cost pricers	0.031	0.037
Log estimated-markup growth: variable-cost pricers	−0.046	0.033
Observations		3,127

Notes: Entries in Panels A and B are survey-weighted percentages. Differences are reported in percentage points. The sample contains firms in France, Hungary, Italy, Spain, and the United Kingdom with a valid E11 response and a valid 2008–2009 estimated markup change. The materials margin is one minus the materials expenditure share. In Panel B, a reported non-increase combines firms reporting a decrease or no change. Panel C reports coefficients from survey-weighted linear probability models in which the dependent variable equals one if the firm reports a margin increase. The specifications for total-cost and variable-cost pricers contain 2,435 and 692 observations, respectively.

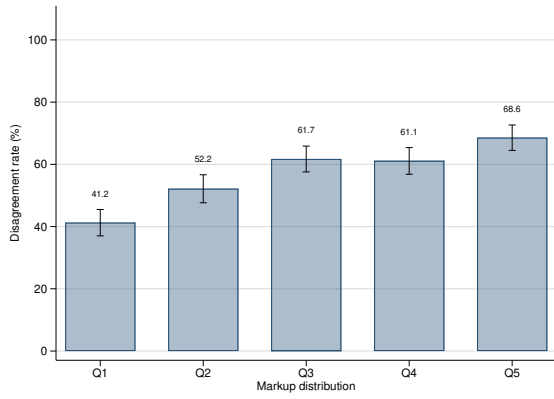
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.4 The Scope of the Disagreement

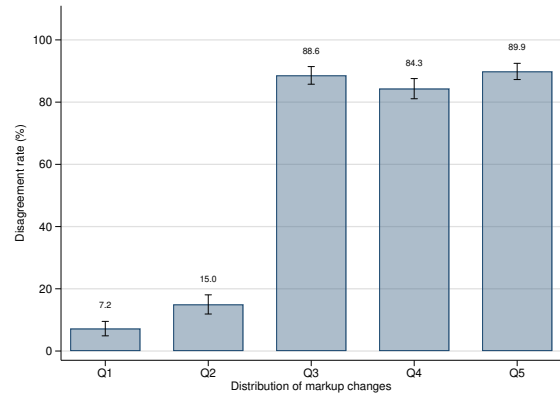
The preceding results show that reported and estimated changes frequently point in different directions. This disagreement would be less consequential if it reflected small estimated changes that happened to fall on opposite sides of zero. It would also be easier to address if it were concentrated among a particular class of firms, such as firms with low estimated markups or firms that account for little economic activity. I examine these possibilities in Figure 1.

Panel (a) reports disagreement rates across the estimated markup distribution. The rate increases from 41.2% in the first quintile to 68.6% in the fifth. The disagreement

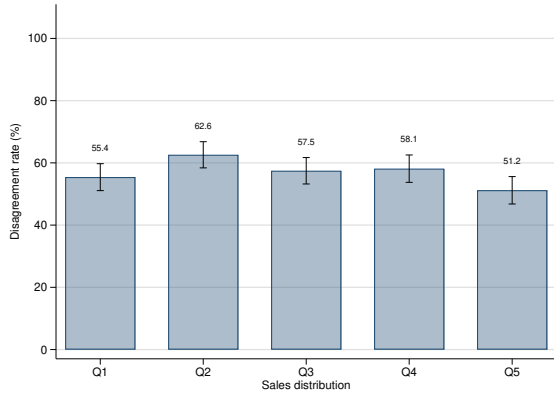
Figure 1: Disagreement Along the Distribution



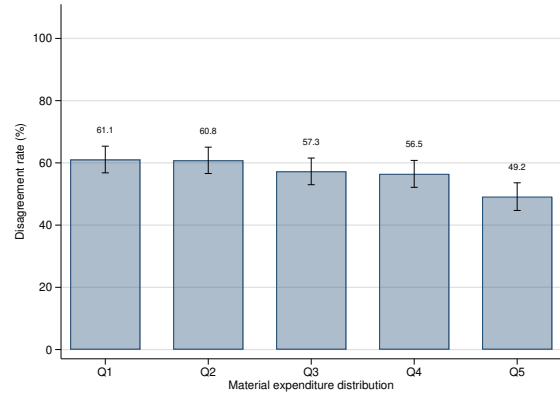
(a) Markup level



(b) Markup change



(c) Sales



(d) Material expenditure

Notes: Each panel divides firms into survey-weighted quintiles of the indicated variable and reports the survey-weighted share of firms for which the reported margin direction and the estimated markup direction disagree. Whiskers report 95% confidence intervals. The sample contains 3,127 firms in France, Hungary, Italy, Spain, and the United Kingdom.

is therefore not confined to estimates close to a markup of one. If anything, it is most prevalent among firms to which the production approach assigns the highest markups.

Panel (b) orders firms by their estimated markup change. Disagreement is 7.2% and 15% in the first two quintiles, but rises to 88.6% in the third, 84.3% in the fourth, and 89.9% in the fifth. The firms receiving the largest estimated markup increases are consequently among those for which the survey is most likely to report a different direction. The disagreement is not a classification problem generated by small changes around zero. It extends to the estimated increases that would provide the strongest evidence of rising market power.

Panels (c) and (d) examine whether the result is concentrated among economically small firms. Disagreement remains between 51.2% and 62.6% across sales quintiles and between 49.2% and 61.1% across material-expenditure quintiles. In the top quintile,

the disagreement rate is 51.2% when firms are ranked by sales and 49.2% when they are ranked by material expenditure. Weighting observations directly by sales or material expenditure produces disagreement rates of 51.9% and 50.5%, respectively. The disagreement therefore remains present among firms that receive substantial weight in aggregate statistics.

These patterns do not identify the source of the disagreement. They do, however, rule out two comparatively innocuous interpretations of the main result. The disagreement cannot be eliminated by discarding estimated changes close to zero, nor does it appear to originate from a readily identifiable group of small firms. It instead appears to be a broader feature of production-based markup dynamics in this setting. Appendix Table B.2 confirms this conclusion after excluding progressively larger changes around zero and under alternative production function specifications.

4 Can Pricing Strategies Explain the Disagreement?

The EFIGE survey distinguishes between firms according to the pricing strategy they follow. Some firms set prices by applying a margin to total costs, while others apply a margin to variable costs. This distinction may help explain why reported margins and production-based markups evolve differently. Variable-cost pricing is conceptually closer to the flexible-input approach, whereas total-cost pricing incorporates expenditures that do not enter the marginal-cost condition used to estimate markups.

The distinction produces two empirical tests. First, reported and estimated changes should correspond more closely among firms pricing over variable costs. Second, among firms pricing over total costs, fixed cost absorption may cause average profitability to fall even when the markup over marginal cost increases. I examine these two possibilities in turn.

4.1 Total- and Variable-Cost Pricing

Question E10 asks firms whether they set prices by applying a margin to total or variable costs. In the final comparison sample, 78.2% of firms report pricing over total costs, while 21.8% report pricing over variable costs. The latter provide the closest comparison with the production approach because fixed costs do not enter their stated pricing strategy.

Variable-cost pricing does not necessarily imply that firms price over marginal cost. Variable costs may include labor, energy, distribution, and other expenditures in addition to materials, while average variable cost need not coincide with marginal cost. Nevertheless, if the disagreement documented above arises primarily because most firms price over total costs, it should become smaller among firms explicitly reporting a variable-cost

pricing strategy.

The data do not support this implication. Among variable-cost pricers, 9.9% report a margin increase, while estimated markups increase for 62.6%, producing a difference of 52.7 percentage points. Among total-cost pricers, 11.4% report a margin increase, while estimated markups increase for 60.5%. The difference is 49.1 percentage points. At the firm level, the reported and estimated directions differ for 59.7% of variable-cost pricers and 56.2% of total-cost pricers. The difference between these rates is 3.5 percentage points and is not statistically significant.

The conditional comparison leads to the same conclusion. Among variable-cost pricers, estimated markups increase for 64.8% of firms reporting a margin increase and for 62.4% of firms reporting a non-increase. Their difference is only 2.4 percentage points and is statistically insignificant. Among total-cost pricers, the corresponding rates are 68.8% and 59.4%, with a statistically significant difference of 9.4 percentage points. The two conditional relationships cannot themselves be statistically distinguished, but there is no evidence that variable-cost pricing strengthens the correspondence between reported and estimated changes.

The magnitude of the estimated change is similarly uninformative. The coefficient on log markup growth in Equation (12) is 0.031 with a standard error of 0.037 among total-cost pricers and -0.046 with a standard error of 0.033 among variable-cost pricers. In neither group are firms receiving larger estimated markup increases more likely to report an increase in their margins.

The disagreement is therefore not explained by the composition of firms across the two pricing strategies. In particular, it is not concentrated among firms pricing over total costs. If anything, the comparison is weaker among firms whose stated pricing strategy is closest to the flexible-input approach.

4.2 Fixed Costs and Average Profitability

The comparison between pricing strategies does not support the view that total-cost pricing alone explains the disagreement. Fixed costs may nevertheless affect the margins reported by firms following this strategy. A firm can experience an increase in its markup over marginal cost while its average profitability falls because fixed costs are spread over lower revenues. I examine this possibility among total-cost pricers.

The relationship derived in Equation (5) identifies fixed cost absorption as an additional source of disagreement between reported margins and marginal-cost markups. To quantify this channel, I consider the benchmark in which marginal and average variable costs coincide, $MC_{it} = AVC_{it}$ as in De Ridder (2024). This restriction allows marginal cost to change over time, but implies that variable production costs equal $MC_{it}Q_{it} = R_{it}/\mu_{it}$.

Letting π_{it}^{op} denote operating profits, the average profit margin therefore satisfies

$$m_{it}^{\pi} = \frac{\pi_{it}^{op}}{R_{it}} = \left(1 - \frac{1}{\mu_{it}}\right) - \frac{F_{it}}{R_{it}}. \quad (13)$$

Fixed operating expenditures do not enter the flexible-input first-order condition directly, but they reduce the share of revenues remaining after the firm has covered its operating costs. Consequently,

$$\Delta m_i^{\pi} \leq 0 \iff \Delta \left(\frac{F_i}{R_i}\right) \geq \Delta \left(1 - \frac{1}{\mu_i}\right). \quad (14)$$

A stable level of fixed costs can therefore cause average profitability to decline when revenues fall, even if the markup increases.

Following De Ridder (2024), I recover the fixed cost share from the difference between the marginal-cost margin implied by the estimated markup and a constructed operating-profit margin:

$$\frac{\widehat{F}_{it}}{R_{it}} = \left(1 - \frac{1}{\widehat{\mu}_{it}}\right) - \frac{\widehat{\pi}_{it}^{op}}{R_{it}}, \quad \widehat{\pi}_{it}^{op} = R_{it} - W_{it} - M_{it} - D_{it}, \quad (15)$$

where W_{it} denotes employee costs, M_{it} materials expenditure, and D_{it} depreciation. I use this resulting operating-profit margin as an accounting proxy for the margin over total costs and set negative fixed costs estimates to zero. I recover each firm's fixed cost level in 2008, hold it constant in real terms, and calculate

$$\widetilde{m}_{it} = 1 - \frac{1}{\widehat{\mu}_{it}} - \frac{\widehat{F}_{i,2008}}{R_{it}}, \quad t \in \{2008, 2009\}. \quad (16)$$

Both estimated markups and revenues are allowed to move; the question is whether spreading the fixed costs recovered for 2008 over 2009 revenues reverses the estimated direction. Revenue falls for 82.1% of firms, making this channel particularly relevant over the period considered.

Table 2 reports the results. Estimated markups increase for 60.5% of firms, while only 11.4% report an increase in their margins. Accounting for fixed cost absorption reduces the estimated increase rate to 54.8%, a decline of 5.7 percentage points. The adjusted rate nevertheless remains 43.4 points above the rate reported in the survey.

The adjustment also improves the firm-level disagreement. Among firms reporting a non-increase but receiving an estimated markup increase, fixed cost absorption reverses the estimated direction in 11.4% of cases. Overall agreement rises from 43.9% to 49.3%, leaving just over half of firms classified differently.

The adjusted margin separates the survey responses more clearly. It increases for

Table 2: Fixed costs and average profitability

	Estimate	SE	R^2
<i>Panel A. Incidence and reconciliation</i>			
Reported margin increased	11.38	0.72	
Estimated markup increased	60.45	1.11	
Adjusted margin increased	54.75	1.13	
Adjusted minus reported increase rate	43.37 ^{***}	1.29	
Reduction in the estimated increase rate	5.70 ^{***}	0.55	
Reported and estimated directions agree	43.86	1.12	
Reported and adjusted directions agree	49.32	1.14	
Gain in agreement	5.46 ^{***}	0.55	
Central disagreement reconciled	11.42	0.97	
<i>Panel B. Adjusted increase rates by reported direction</i>			
Reported non-increase	53.06	1.20	
Reported increase	67.89	3.14	
Difference: increase minus non-increase	14.83 ^{***}	3.36	
<i>Panel C. Predicting a reported margin increase</i>			
Standardized unadjusted margin change	0.014 ^{**}	0.007	0.002
Standardized adjusted margin change	0.027 ^{***}	0.007	0.007
Observations			2,415

Notes: Entries are survey-weighted percentages. Differences in Panels A and B are expressed in percentage points. Panel C instead reports coefficients from separate survey-weighted linear probability models, with the dependent variable equal to one for a reported margin increase. The unadjusted margin is $1 - 1/\hat{\mu}_{it}$; the adjusted margin holds the fixed cost level recovered in 2008 constant in real terms. Both explanatory variables in Panel C are changes winsorized at the first and ninety-ninth percentiles and divided by their weighted standard deviations. The central disagreement consists of firms reporting a non-increase while receiving an estimated markup increase. Operating profits are revenues less employee costs, materials expenditure, and depreciation. The sample contains the 2,435 total-cost pricers in the comparison sample, excluding 20 firms with missing depreciation information required to recover fixed costs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

67.9% of firms reporting a higher margin and for 53.1% of firms reporting a non-increase, a difference of 14.8 percentage points. To examine the continuous relationship, I compare changes in the unadjusted marginal-cost margin, $1 - 1/\hat{\mu}_{it}$, with changes in its adjusted counterpart. A one-standard-deviation increase in the adjusted margin is associated with a 2.7-percentage-point higher probability of reporting a margin increase, compared with 1.4 points for the unadjusted margin. Explanatory power rises from 0.2% to 0.7%, but remains limited.

Under this benchmark, absorbing the fixed costs recovered for 2008 shrinks the gap between reported and estimated changes but leaves most of it unexplained. Fixed costs therefore matter for the comparison, although the absorption channel examined here does not account for the main empirical result.

5 Conclusion

This paper asks whether production-based markup estimates recover changes in firms' price-cost margins. The EFIGE database provides a unique opportunity to examine this question by linking firms' financial information to survey responses on their pricing strategies and on the evolution of the margins applied to their costs. I compare the change reported by each firm with the change in its production-based markup over the same period.

The two measures provide substantially different accounts of firms' margin dynamics. Only 11% of firms report that their margin increased between 2008 and 2009, while estimated markups increase for 60.9%. The production approach therefore assigns increases to a share of firms almost 50 percentage points larger than that reported in the survey. Reported and estimated directions agree for only 43% of firms, and the magnitude of estimated markup growth is essentially unrelated to the survey response in the linear regressions. The disagreement is not confined to small estimated movements around zero or to a particular segment of the distributions of markups, markup changes, sales, or materials expenditure.

The distinction between pricing strategies does not resolve this disagreement. Among firms explicitly pricing over variable costs, estimated markups increase for 62.6%, while only 9.9% report a margin increase. The comparison therefore does not improve among firms whose stated pricing strategy excludes fixed costs, thus aligning more closely with the markup definition. Among total-cost pricers, the fixed cost adjustment reduces the gap only modestly. Under the fixed cost benchmark, the incidence of increases falls from 60.5% for estimated markups to 54.8% for adjusted margins, while agreement with the survey rises from 43.9% to 49.3%. The adjusted increase rate nevertheless remains 43.4 percentage points above the reported rate.

The survey does not define the expenditures included in total or variable costs, and the mapping between reported margins and marginal-cost markups depends on the relationship between average and marginal costs and, for total-cost pricers, on fixed cost absorption. The survey nevertheless provides information about a margin explicitly connected to the firm's pricing rule and elicited independently of the production function estimation. The evidence establishes a substantial disagreement between these independently constructed measures of firm-level changes, which persists across pricing strategies and after the fixed cost adjustment considered here.

Markups are not directly observed, and robustness across production function specifications, or agreement between estimates constructed from the same first-order condition, cannot by itself establish that their estimated changes recover the underlying economic object. The evidence in this paper indicates that annual production-based markup changes do not track the margins firms report. Further validation against independently

collected information on firms' pricing and costs is therefore essential before interpreting these estimated changes as evidence of the evolution of market power.

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Appendix

A Data Preparation

A.1 Sample construction

The production functions are estimated using all firm-year observations in the EFIGE data that satisfy the following requirements. Observations must have positive values for sales, employment, tangible fixed assets, and materials expenditure, together with the lagged variables required by the estimation. I first estimate markups and construct annual markup changes using the EFIGE panel. I then merge the survey response and retain firms that report setting prices as a margin over total or variable costs and answer question E11. The resulting comparison sample contains 3,127 firms in France, Hungary, Italy, Spain, and the United Kingdom, with the sampling weights accounting for the stratified design of the original survey.

Appendix Table A.1 reports summary statistics for this sample. Financial variables and employment are measured in 2008, before the change examined in the paper. The average firm has 42 employees, while the median has 22. The difference between average and median sales reflects the right-skewed size distribution of manufacturing firms. The estimated markup has a mean of 1.54 and a median of 1.31.

Table A.1: Summary statistics

	Mean	Median	SD	Observations
Sales	9,114.0	3,392.3	23,505.0	3,127
Materials expenditure	5,064.8	1,485.5	16,578.8	3,127
Tangible fixed assets	2,163.0	603.9	6,695.6	3,127
Employment	41.7	22.0	70.8	3,127
Estimated markup	1.543	1.309	0.997	3,127
Log change in estimated markup	0.037	0.023	0.188	3,127

Notes: Statistics are survey weighted. Sales, materials expenditure, tangible fixed assets, employment, and the estimated markup are measured in 2008. Financial variables are expressed in thousands of euros in common prices. The markup change is the log change between 2008 and 2009.

A.2 Additional country statistics

Table A.2: Pricing strategies and reported margin changes

Country	Active price setters	Variable-cost pricers	Total-cost pricers	Increased margins
FRA	54.9	28.4	71.6	13.3
HUN	32.8	26.0	74.0	2.7
ITA	56.2	23.7	76.3	13.6
SPA	66.8	15.6	84.4	6.6
UK	60.6	37.7	62.3	20.4
Five-country sample	57.9	21.8	78.2	11.0

Notes: Entries are survey-weighted percentages. Active price setters are calculated among firms with a valid response to E10 in the five countries used in the analysis.

Table A.3: Reported and estimated changes by country

<i>Panel A. Reported and estimated directions</i>			
Country	Reported non-increase	Estimated non-increase	Gap
FRA	86.67	42.16	44.52
HUN	97.26	53.43	43.83
ITA	86.39	37.37	49.01
SPA	93.39	38.69	54.70
UK	79.59	47.75	31.84
<i>Panel B. Firm-level discrepancies</i>			
Country	Reported non-inc. Estimated inc.	Reported inc. Estimated non-inc.	Restricted-subset increase
FRA	48.46	3.95	53.14
HUN	46.57	2.74	36.87
ITA	53.43	4.42	68.89
SPA	56.66	1.97	67.76
UK	39.72	7.88	61.18

Notes: Entries are survey-weighted percentages. Differences are reported in percentage points. Panel A compares the reported and estimated non-increase rates. “Gap” is the reported margin non-increase rate minus the estimated markup non-increase rate, which is equivalent to the estimated increase rate minus the reported increase rate. Panel B reports the two firm-level disagreement cells. “Restricted-subset increase” is the estimated markup increase rate among firms reporting a margin non-increase whose estimated materials elasticity does not increase.

B Production Function Estimation and Robustness

B.1 Estimation

The production function in Equation (8) is estimated separately for each country. Expanding the translog specification gives

$$\begin{aligned} f_c(k_{it}, l_{it}, x_{it}; \beta_c) = & \beta_c^K k_{it} + \beta_c^L l_{it} + \beta_c^M x_{it} + \beta_c^{KK} k_{it}^2 + \beta_c^{LL} l_{it}^2 + \beta_c^{MM} x_{it}^2 \\ & + \beta_c^{KL} k_{it} l_{it} + \beta_c^{KM} k_{it} x_{it} + \beta_c^{LM} l_{it} x_{it}. \end{aligned} \quad (\text{B.1})$$

The first stage estimates expected revenue using a third-order polynomial in current capital, labor, and materials, together with year effects and the firm's survey-weighted share of sales within its country, two-digit industry, and year. Denoting the resulting fitted value net of the transitory shock by $\hat{\phi}_{it}$, productivity for any candidate parameter vector is

$$\omega_{it}(\beta_c) = \hat{\phi}_{it} - f_c(k_{it}, l_{it}, x_{it}; \beta_c). \quad (\text{B.2})$$

Productivity follows a first-order Markov process,

$$\omega_{it} = g_c(\omega_{it-1}) + \xi_{it}, \quad (\text{B.3})$$

where $g_c(\cdot)$ is approximated linearly. The parameters are estimated by imposing

$$\mathbb{E} [\xi_{it}(\beta_c) Z_{it}] = 0, \quad (\text{B.4})$$

where Z_{it} contains current capital and lagged labor and materials, together with their second-order and interaction terms. These timing restrictions follow the control-function logic of Olley and Pakes (1996), Levinsohn and Petrin (2003), and Akerberg et al. (2015).

The estimated materials elasticity is obtained from Equation (9). Materials elasticities and markups are winsorized at the second and ninety-ninth percentiles within each country, and economically nonsensical negative estimates are dropped. Annual changes are calculated after these steps. The production functions use all eligible accounting observations; the E11 response is merged only when the firm-level comparison is constructed.

Appendix Table B.1 reports the distribution of estimated materials elasticities across the five countries used in the analysis.

Table B.1: Estimated materials elasticities by country

Country	Mean	Median	SD
FRA	0.478	0.478	0.136
HUN	0.735	0.711	0.167
ITA	0.554	0.556	0.112
SPA	0.580	0.582	0.113
UK	0.727	0.722	0.061

Notes: Statistics are survey weighted.

B.2 Alternative samples and production function specifications

Appendix Table B.2 examines whether the main results depend on the country sample, the timing assigned to the survey response, the classification of small estimated changes, or the assumptions used to estimate the production function.

The first four rows vary the sample and timing. The five-country comparison sample reproduces the specification used in the paper. The next row retains all seven EFIGE countries and therefore tests whether excluding Austria and Germany affects the estimates. Restricting the sample to France, Italy, and Spain examines whether the results are driven by the two smaller country samples retained in the paper. The 2007–2008 specification compares the survey response with the preceding annual change rather than the 2008–2009 change used in the main analysis.

The next two rows remove firms with absolute estimated log markup changes below 0.01 or 0.05. These specifications address the possibility that the comparison is driven by estimated movements close to zero. The incidence difference becomes larger, rather than smaller, as firms with small estimated changes are removed.

The remaining rows modify the production function estimation. The specification without the market-share control removes the firm’s market share from the first-stage control function. The labor-as-state-input specification replaces the moments based on lagged labor with current labor and the corresponding second-order and interaction terms. It therefore considers an alternative timing assumption under which employment is chosen before the within-period productivity innovation.

The industry specification estimates separate production functions within two-digit manufacturing industries, pooling countries and including country-year effects. Production functions are estimated only for industries with at least 500 eligible observations. This allows the production technology and output elasticities to vary across industries rather than primarily across countries. Finally, the overlapping-window specification estimates country-specific production functions using observations from 2004–2008 for the 2008 elasticity and from 2005–2009 for the 2009 elasticity. It therefore allows the production parameters assigned to 2008 and 2009 to differ, rather than imposing

parameter stability over the full accounting panel.

The incidence difference remains large in every specification. It equals 49.6 percentage points when all seven countries are retained and 50.5 points when the sample is restricted to France, Italy, and Spain. Removing estimated changes close to zero raises the difference to 51.3 or 53.4 points. Omitting the market-share control, changing the labor moments, and estimating production functions by industry produce differences between 49.2 and 49.6 percentage points. Using overlapping five-year windows produces an even larger difference of 61.2 points. The conditional difference also remains positive across all four alternative production function specifications. The main result is therefore not specific to the country sample, small estimated movements, or a particular implementation of the production function estimation.

Table B.2: Alternative samples and production function specifications

Specification	Gap	Conditional difference	Restricted-subset increase
Five-country analysis	49.90	7.96***	66.02
All EFIGE countries	49.58	6.86**	65.58
France, Italy, and Spain	50.54	8.13***	66.58
2007–2008 change	40.13	-4.55	60.82
Drop $ \Delta \log \mu < 0.01$	51.34	5.85*	67.89
Drop $ \Delta \log \mu < 0.05$	53.43	11.04***	70.83
No market-share control	49.35	6.57**	65.54
Labor as a state input	49.62	7.41**	65.78
Two-digit industry PFs	49.19	8.32***	64.17
Overlapping five-year PFs	61.17	8.94***	60.05

Notes: Entries are survey-weighted percentages. Differences are reported in percentage points. “Gap” is the estimated markup increase rate minus the reported margin increase rate. “Conditional difference” is the estimated markup increase rate among firms reporting an increase minus the corresponding rate among firms reporting a non-increase, as in Table 1. “Restricted-subset increase” is the estimated markup increase rate among firms reporting a non-increase whose estimated materials elasticity also does not increase. The five-country comparison sample contains France, Hungary, Italy, Spain, and the United Kingdom; the 2007–2008 row retains the same five countries. The production function variants retain every country for which the corresponding specification can be estimated.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 Timing and the continuous comparison

EFIGE fieldwork took place between January and May 2010, and E11 refers to changes “during the last year.” The paper therefore compares 2009 with 2008. Appendix Table B.3 reports results when the same survey response is instead compared with the 2007–2008 change or the two-year change between 2007 and 2009. The incidence difference is large under all three conventions, ranging from 40.1 to 49.9 percentage points. The 2008–2009 comparison is retained because it provides the closest correspondence with

the wording and timing of the survey.

Table B.3: Alternative timing conventions

Timing window	Gap	Survey non-inc. Markup inc.	Survey inc. Markup non-inc.	Subset increase
2007–2008	40.13	45.79	5.67	60.82
2008–2009	49.90	53.43	3.53	66.02
2007–2009	48.34	52.19	3.85	63.63

Notes: Entries are survey-weighted percentages. Differences are reported in percentage points. All rows retain the five countries used in the analysis. The survey response is unchanged across rows. “Gap” and “subset increase” are defined as in Appendix Table A.3, using estimated changes over the indicated accounting window.

Appendix Table B.4 extends the continuous comparison in Equation (12). Adding country fixed effects, two-digit industry fixed effects, log sales, and log employment has virtually no effect on the estimate. In every specification, estimated markup growth has no economically or statistically significant relationship with the survey response.

Table B.4: Estimated markup growth and reported margin changes

	(1)	(2)	(3)
Estimated markup growth (log change)	0.009 (0.029)	0.014 (0.029)	-0.005 (0.029)
Country fixed effects	No	Yes	Yes
Sector fixed effects	No	No	Yes
Size controls	No	No	Yes
Observations	3075	3075	3075

Notes: Survey-weighted linear probability models. All columns use the same 3,075 firms from the main five-country comparison sample with complete controls. The dependent variable equals one if the firm reports that its margin increased, as in Equation (12). Size controls are log sales and log employment. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.